

All-Orders Endpoint Asymptotics for the Inverse Fabius Function

**Diagonal Lagrange–Bürmann coefficients, saddle partition sums,
phase-locked tomography, inverse derivatives,
and the exact 2-adic completion of the Laplace exponent**

Consolidated frontier volume for Vladimir Reshetnikov’s ProveIt project

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Abstract

Let F be the Fabius distribution function and $G = F^{-1}$ its quantile. The corpus proves an all-orders small- x expansion of $\log F$ in the exact lower-Lambert phase, with a one-periodic Gamma–zeta fluctuation Ψ and periodic saddle coefficients A_j , and its inverse volume derives the first two endpoint corrections of G together with a triangular all-orders recursion. This volume consolidates two independently written reports that close the remaining coefficient-level gaps — by, in several places, proving the same theorems, which are stated here once with the merged proofs and cross-validated constants.

With $T = \log(1/y)$, $L = \log 2$, $\rho = \sqrt{2T/L}$, $\ell = \log \rho$, $\beta = 1/L + \frac{1}{2}$, and the fast phase $\theta = \rho + \beta$, the quantile satisfies, for every fixed N ,

$$G(y) = G_0(y) \left(1 + \sum_{n=1}^N \frac{g_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N((1 + \ell)^{N+1} \rho^{-N-1}) \right), \quad G_0(y) = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}},$$

with coefficients polynomial in ℓ and real-analytic one-periodic in θ . A catalytic-variable Lagrange–Bürmann inversion gives every phase, logarithmic, and multiplicative coefficient by one finite nonrecursive extraction; a direct partial-Bell formula gives g_n straight from the phase coefficients. Structure laws follow: the universal highest logarithm $[\ell^n]g_n = 1/(2^n n!)$; the newest-jet transfer $[\Psi^{(2n-2)}]g_n = (-1)^n/(2^{n-1}(n-1)! L^{2n-2})$; explicit Gamma–zeta convolutions for every Fourier mode with a common analyticity strip $|\Im \theta| < \pi/(2L)$; phase-locked quantile sequences and coefficient tomography; an all-orders elasticity recurrence; and, for each fixed m , the two-term derivative law with harmonic-number correction. The explicit third correction lowers the relative error on exact dyadic quantiles $y_n = F(2^{-n})$ to 4.9×10^{-8} at $n = 160$.

On the forward side, the complete Gaussian saddle correction is a finite constrained partition sum (including the Bromwich $1/t$ factor) that specializes to the Mills-ratio coefficients — completing an explicit pipeline $\Psi \rightarrow \{\varkappa_r\} \rightarrow \{C_n\} \rightarrow \{\Lambda_n\} \rightarrow \{A_j\} \rightarrow \{d_n\} \rightarrow \{g_n\}$. And the Laplace exponent $K = \log \mathbb{E} e^{-tX}$ admits the *exact* decomposition

$$K(t) = -\frac{\log^2 t}{2L} + \frac{1}{2} \log t - C_B + \Psi(\log_2 t) + B(t), \quad B(t) = \sum_{n \geq 1} \frac{2^{\nu_2(n)+1} - 1}{n} e^{-nt},$$

whose 2-adically weighted Bose tail has Mahler equation, Dirichlet series $\zeta(s+1)/(1-2^{-s})$, and a Mellin bridge showing that the periodic phase and the exponentially small dyadic sectors are the two faces of one kernel — the concrete seed of an exponentially improved inverse theory. Provenance of the absorbed sources, with SHA-256 hashes and the cross-validation record, is in [section C](#).

Contents

Provenance and merge conventions	3
1 Imported inputs and the two-scale inverse problem	3
1.1 Forward input	3
1.2 Inverse prior art	4
1.3 The two-scale problem	4
2 The diagonal Lagrange–Bürmann calculus	5
3 The explicit third correction	6
4 Structure laws	6
4.1 Logarithmic degrees and top logarithms	7
4.2 Transfer of the newest periodic derivative	7
5 Fourier-mode algebra and the analyticity strip	8
6 Phase locking and coefficient tomography	8
7 Elasticity and every inverse derivative	8
8 The forward saddle engine: constrained partition sums	9
9 The exact 2-adic completion of the Laplace exponent	10
9.1 The decomposition and the dyadic Bose tail	10
9.2 One Mellin kernel, two asymptotic ends	11
9.3 Exact saddle map and ultraflat jets	12
10 Verification	12
11 Connections across the corpus	14
12 Conjectures and frontier directions	14
A Proof details	15
B Coefficient dictionary	15
C Provenance of the absorbed sources	16

Provenance and merge conventions

This volume is an editorial consolidation (28 August 2026) of two drafts of the fourth and fifth waves that attack the same problem and overlap heavily: both prove the closed Lagrange–Bürmann coefficient formula, the universal highest logarithm, the explicit third correction, and the all-orders elasticity recurrence. Each shared theorem is stated once; where the sources’ formulas could be compared they were verified to agree exactly (notably the third correction d_3 , derived independently in different groupings, and the elasticity recurrence); and each source’s unique material — the structure laws, Fourier algebra, tomography, and derivative hierarchy of one, the saddle partition engine and the 2-adic completion of the other — keeps its own chapter. [Section C](#) records the sources with SHA-256 hashes and the deduplication detail; supporting files are under `assets/`.

Status discipline

Theorems have complete proofs here or explicit citations; the repository inputs (the forward all-orders expansion, Ψ , $C_\sharp$, A_1 , the forward top-jet law, and the first two inverse corrections with the triangular engine of the *Inverse and Sampling Frontiers* volume) are imported, not reproved. “New” means absent from the audited corpus snapshots under the normalizations used there; no unqualified worldwide priority is claimed. Conjectures and problems are labeled; nothing here is formalized in Lean yet.

Notation

$F, G = F^{-1}$	Fabius function and quantile	L	$\log 2$
β	$1/L + \frac{1}{2}$	T	$\log(1/y)$
ρ	$\sqrt{2T/L}$; $z = 1/\rho$	ℓ	$\log \rho$
θ	$\rho + \beta \pmod{1}$ in periodic functions	$\lambda(x)$	$-W_{-1}(-Lx)/L$, so $x = \lambda e^{-L\lambda}$
Ψ	mean-zero periodic Gamma–zeta fluctuation	A_j	periodic forward saddle coefficients
D	$\lambda(G(y)) - \rho - \beta = \sum d_n z^n$	h_n, g_n	coefficients of $\log(G/G_0)$, G/G_0
$K(t)$	$\log \mathbb{E} e^{-tX}$ (Laplace exponent)	$B(t)$	dyadic Bose tail (9.3)
$\varkappa_r$	$s^r K^{(r)}(s)$ at the saddle	ϱ_r	$\varkappa_r / \varkappa_2$
$Y_n, Y_{n,k}$	complete/partial exponential Bell	H_r	harmonic number $\sum_{j \leq r} 1/j$

The constants $C_\sharp = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi$ and $C_B = \frac{L}{12} + \frac{\pi^2 - 6\gamma^2 - 12\gamma_1}{12L}$ (Euler γ , first Stieltjes γ_1) satisfy the compact relation $C_\sharp + C_B = -\frac{1}{2} \log(2\pi)$, proved with the 2-adic completion in [section 9](#).

1 Imported inputs and the two-scale inverse problem

1.1 Forward input

With $X = \sum_{j \geq 1} 2^{-j} U_j$ (U_j i.i.d. uniform on $[0, 1]$) and $F(x) = \mathbb{P}(X \leq x)$, the Laplace transform and exponent are

$$M(t) = \mathbb{E} e^{-tX} = \prod_{j \geq 1} \frac{1 - e^{-t/2^j}}{t/2^j}, \quad K(t) = \log M(t), \quad (1.1)$$

and the Bromwich integral $F(x) = \frac{1}{2\pi i} \int_{(\sigma)} t^{-1} \exp(xt + K(t)) dt$ is the analytic source of the saddle expansion. For small $x > 0$ the exact lower-Lambert phase is $\lambda(x) = -W_{-1}(-Lx)/L$, i.e. $x = \lambda e^{-L\lambda}$ exactly, and the repository’s forward theorem states: for every fixed $N \geq 1$,

$$\log F(x) = -\frac{L}{2} \lambda^2 + L\beta\lambda - \frac{1}{2} \log \lambda + C_\sharp + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}_N(\lambda^{-N}), \quad (1.2)$$

uniformly in the phase, with every A_j a bounded smooth one-periodic differential polynomial in the Gamma-zeta fluctuation

$$\Psi(u) = \sum_{k \neq 0} \widehat{\Psi}(k) e^{2\pi i k u}, \quad \widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}, \quad (1.3)$$

$\widehat{\Psi}(0) = 0$. The explicit first correction and the forward top-jet law are

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}, \quad [\Psi^{(2j)}]A_j = \frac{(-1)^j}{2^j j! L^{2j}}. \quad (1.4)$$

The second correction, needed for the third inverse order, is

$$A_2 = \frac{P_1}{24L} + \frac{2P_1 + 1}{4L^2} - \frac{P_1^3 + 3P_1^2 + 3P_1P_2 + 3P_2 + P_3}{6L^3} + \frac{4P_1^2P_2 + 4P_1P_3 + 2P_2^2 + P_4}{8L^4}, \quad (1.5)$$

with $P_r = \Psi^{(r)}(\theta)$, and $A'_1 = -(P_3 + 2P_1P_2)/(2L^2)$.

1.2 Inverse prior art

The corpus's inverse volume establishes, and this volume imports: the leading factor and first two phase corrections

$$d_1 = \frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta) - \frac{1}{2}\ell}{L}, \quad d_2 = \frac{\Psi'(\theta) d_1 + A_1(\theta) - \frac{1}{2}\beta}{L}; \quad (1.6)$$

the logarithmic corrections $h_1 = \frac{1}{2}\ell + \kappa_0 - \Psi(\theta)$ with $\kappa_0 = \beta - \frac{L}{2}\beta^2 - C_\#$, and h_2 ; the nondegenerate first normalized cluster interval; the two-term elasticity; and a triangular all-orders recursion. The consolidated results below convert that recursion into closed formulas and extend every structural layer.

1.3 The two-scale problem

For $y \downarrow 0$ set T, ρ, z, ℓ, θ as in the notation table and

$$G_0(y) = \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}, \quad \lambda(G(y)) = \rho + \beta + D, \quad D = \sum_{n \geq 1} d_n z^n. \quad (1.7)$$

Substituting $\lambda = \rho + \beta + D$ into (1.2) and using $-\frac{L}{2}\lambda^2 + L\beta\lambda = -T - L\rho D + \frac{L}{2}\beta^2 - \frac{L}{2}D^2$ (the quadratic collapses exactly because $T = L\rho^2/2$) reduces $\log F(G(y)) = -T$ to the fixed-point equation

$$\boxed{D = z \Phi(z, D)}, \quad \Phi(z, w) = -\frac{1}{L} \mathcal{R}(z, w), \quad (1.8)$$

with the formal residual

$$\mathcal{R}(z, w) = -\frac{L}{2}\beta^2 + \frac{L}{2}w^2 + \frac{1}{2}\ell + \frac{1}{2}\log(1 + \beta z + zw) - C_\# - \Psi(\theta + w) - \sum_{j \geq 1} \frac{A_j(\theta + w) z^j}{(1 + \beta z + zw)^j}. \quad (1.9)$$

Coefficients are computed in the two-scale ring $\mathcal{A}[\ell][[z]]$, where $\mathcal{A}$ is the differential algebra of one-periodic functions generated by Ψ and the A_j : ℓ and θ are treated as independent coefficient variables and only afterwards evaluated at $\log \rho$ and $\rho + \beta$. This convention is forced, not cosmetic: θ winds around the circle and ℓ is unbounded, so expanding either as a slow series would destroy terms at their own order — there is no phase-free complete expansion (section 4), and the correct coefficient space is $\mathbb{R}[\ell] \otimes C^\infty(\mathbb{R}/\mathbb{Z})$.

Theorem 1.1 (All-orders inverse endpoint expansion). *For every fixed $N \geq 1$ there are unique coefficients $d_n, h_n, g_n \in \mathcal{A}[\ell]$ ($1 \leq n \leq N$), polynomials in ℓ with real-analytic one-periodic coefficients, such that as $y \downarrow 0$, uniformly in the phase,*

$$\lambda(G(y)) = \rho + \beta + \sum_{n=1}^N \frac{d_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^N}{\rho^{N+1}}\right), \quad (1.10)$$

$$\log \frac{G(y)}{G_0(y)} = \sum_{n=1}^N \frac{h_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^N}{\rho^{N+1}}\right), \quad \frac{G(y)}{G_0(y)} = 1 + \sum_{n=1}^N \frac{g_n(\ell, \theta)}{\rho^n} + \mathcal{O}_N\left(\frac{(1+\ell)^{N+1}}{\rho^{N+1}}\right). \quad (1.11)$$

d_n depends only on $\Psi, A_1, \dots, A_{n-1}$ and finitely many of their derivatives. By reflection, $G(1-y) = 1 - G(y)$ transfers every formula to the right endpoint.

Proof. Substituting the truncation $D^{[N]} = \sum_{n \leq N} d_n z^n$ into (1.8) and cancelling $z, \dots, z^N$ determines the d_n uniquely and triangularly (D enters multiplied by z); polynomial dependence on ℓ follows because the only explicit ℓ in Φ is affine, and periodic analyticity from closure of $\mathcal{A}$ (section 5). The residual at $D^{[N]}$ is $\mathcal{O}_N((1+\ell)^N z^N)$, while on the connecting segment $\partial_w(\frac{L}{z}w + \mathcal{R}) = \frac{L}{z} + \mathcal{O}_N(1)$; the mean-value theorem gains one power of z and proves (1.10) (details in section A). The exact Lambert identity gives

$$\frac{G}{G_0} = (1 + \beta z + zD) e^{-LD}, \quad H := \log \frac{G}{G_0} = \log(1 + \beta z + zD) - LD, \quad (1.12)$$

and formal composition and exponentiation give (1.11), the extra $(1+\ell)$ bounding the first omitted complete Bell polynomial. $\square$

2 The diagonal Lagrange–Bürmann calculus

The triangular recursion computes order n only after all lower orders. A catalytic variable turns the same fixed point into a closed finite extraction at any single prescribed order — the central shared theorem of the two absorbed sources.

Theorem 2.1 (Diagonal coefficient formulas). *For every $n \geq 1$,*

$$d_n = \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] \Phi(z, w)^k = \sum_{k=1}^n \frac{(-1)^k}{k L^k} [z^{n-k} w^{k-1}] \mathcal{R}(z, w)^k, \quad (2.1)$$

a finite formula: only the z^{n-1} -truncation of Φ , the jets $\Psi, \dots, \Psi^{(n-1)}$, and $A_1, \dots, A_{n-1}$ enter. More generally, for any formal $K(z, w)$,

$$[z^n] K(z, D(z)) = [z^n] K(z, 0) + \sum_{k=1}^n \frac{1}{k} [z^{n-k} w^{k-1}] \partial_w K(z, w) \Phi(z, w)^k, \quad (2.2)$$

which with $K = \log(1 + \beta z + zw) - Lw$ and $K = (1 + \beta z + zw)e^{-Lw}$ gives closed diagonal formulas for h_n and g_n directly.

Proof. Let $W(t, z)$ solve $W = t \Phi(z, W)$. For fixed z , ordinary Lagrange–Bürmann inversion in t gives $[t^k] W = \frac{1}{k} [w^{k-1}] \Phi(z, w)^k$ and the composition form $K(z, W) = K(z, 0) + \sum_k \frac{t^k}{k} [w^{k-1}] \partial_w K \cdot \Phi^k$. Since $D(z) = W(z, z)$, set $t = z$ and extract total degree n . Finiteness: a term with Lagrange index k asks for $z^{n-k} w^{k-1}$, so no $A_j z^j$ with $j > n - k$ and at most $k - 1$ phase derivatives occur. $\square$

Expanding $\Phi = \sum \phi_{r,s} z^r w^s$ gives the fully unrolled partition form $d_n = \sum_k \frac{1}{k} \sum_{r_1+\dots+r_k=n-k, s_1+\dots+s_k=k-1} \prod_j \phi_{r_j, s_j}$ — convenient for proofs, while (2.1) is convenient for implementation.

Theorem 2.2 (Direct partial-Bell formula for g_n). *With $x_j = j! d_j$ and the partial Bell normalization $(\sum_j d_j z^j)^k = \sum_{n \geq k} \frac{k!}{n!} Y_{n,k}(x_1, x_2, \dots) z^n$,*

$$g_n = \frac{1}{n!} \sum_{k=1}^n (-L)^k Y_{n,k} + \frac{\beta}{(n-1)!} \sum_{k=0}^{n-1} (-L)^k Y_{n-1,k} + \frac{1}{(n-1)!} \sum_{k=1}^{n-1} k (-L)^{k-1} Y_{n-1,k}, \quad (2.3)$$

all $Y_{m,k} = Y_{m,k}(x_1, x_2, \dots)$; combined with (2.1) this is a nonrecursive finite formula for the arbitrary-order multiplicative coefficient. The logarithmic-to-multiplicative conversion is the complete-Bell identity $g_n = \frac{1}{n!} Y_n(1!h_1, \dots, n!h_n)$, i.e. $ng_n = \sum_{j \leq n} j h_j g_{n-j}$.

Proof. Expand the exact factorization (1.12): the three lines are $[z^n]e^{-LD}$, $[z^{n-1}]\beta e^{-LD}$, and $[z^{n-1}]De^{-LD}$, each a standard partial-Bell expansion. $\square$

Algorithm 2.3 (Arbitrary-order symbolic generation). Fix an output order N : obtain $A_1, \dots, A_{N-1}$ (from (1.4)–(1.5) or the saddle pipeline of section 8); Taylor-expand $\Psi(\theta + w)$, $A_j(\theta + w)$ only as far as the bidegrees require; truncate Φ at z^{N-1} ; evaluate (2.1) for each needed n (no lower d_j is mathematically required); and produce g_n by (2.3) or the h -route. The triangular and diagonal algorithms are complementary: the former reuses lower-order work, the latter is an independent certificate – the absorbed sources’ implementations verify $d_j^{\text{diag}} \equiv d_j^{\text{tri}}$ symbolically for $j \leq 3$ over algebraically independent jet symbols.

3 The explicit third correction

Theorem 3.1 (Closed third-order coefficients). *With $P_r = \Psi^{(r)}(\theta)$ and d_1, d_2 from (1.6),*

$$d_3 = \frac{1}{L} \left(P_1 d_2 + \frac{1}{2} P_2 d_1^2 + A_1' d_1 - \beta A_1 + A_2 - \frac{L}{2} d_1^2 - \frac{1}{2} d_1 + \frac{\beta^2}{4} \right), \quad (3.1)$$

and

$$h_1 = \beta - L d_1, \quad h_2 = d_1 - \frac{\beta^2}{2} - L d_2, \quad h_3 = d_2 - \beta d_1 + \frac{\beta^3}{3} - L d_3, \quad (3.2)$$

$$g_1 = h_1, \quad g_2 = h_2 + \frac{1}{2} h_1^2 = d_1 - L d_2 - \beta L d_1 + \frac{L^2}{2} d_1^2, \quad g_3 = h_3 + h_1 h_2 + \frac{1}{6} h_1^3, \quad (3.3)$$

with the direct d -form $g_3 = -L d_3 + (1 - \beta L) d_2 + L^2 d_1 d_2 + (\frac{\beta L^2}{2} - L) d_1^2 - \frac{L^3}{6} d_1^3$.

Proof. Extract $[z^2]$ from (1.8): the contributions are $\frac{L}{2} d_1^2, \frac{1}{2} d_1 - \frac{\beta^2}{4}$ (from the logarithm), $-P_1 d_2 - \frac{1}{2} P_2 d_1^2$ (Taylor transport of Ψ), $-A_1' d_1 + \beta A_1$ (transport of A_1), and $-A_2$; their sum plus $L d_3$ vanishes. The h - and g -chains are the expansions of (1.12). $\square$

Remark 3.2 (Cross-validation of the sources). The two absorbed reports derived (3.1) independently and grouped it differently (one as $-\frac{1}{L} \{ \frac{L}{2} d_1^2 + \frac{1}{2} d_1 - \frac{\beta^2}{4} - P_1 d_2 - \frac{P_2}{2} d_1^2 - A_1' d_1 + \beta A_1 - A_2 \}$); the forms are algebraically identical, and both were checked symbolically against the triangular recursion in their independent implementations. The compact grouping displays the five mechanisms separately – quadratic saddle correction, logarithmic Jacobian, transport of Ψ , transport of A_1 , and the genuinely new A_2 – and is the stabler form for evaluation.

4 Structure laws

The diagonal formula does more than compute: its bidegree geometry yields laws invisible in a triangular expansion.

4.1 Logarithmic degrees and top logarithms

Write $\mathcal{R} = \frac{1}{2}\ell + \mathcal{R}_0$ with $\mathcal{R}_0$ independent of ℓ , so the z^0 slice of Φ is $\Phi(0, w) = a\ell + \phi(w)$ with

$$a = -\frac{1}{2L}, \quad \phi(w) = \frac{\beta^2}{2} + \frac{C_\# + \Psi(\theta + w)}{L} - \frac{w^2}{2}. \quad (4.1)$$

Theorem 4.1 (Degree and top-log laws). $\deg_\ell d_1 = 1$ and $\deg_\ell d_n \leq n - 1$ for $n \geq 2$, with

$$[\ell^{n-1}]d_n = a^{n-1} [w^{n-1}] \phi(w) \quad (n \geq 2); \quad (4.2)$$

explicitly $[\ell]d_2 = -\Psi'(\theta)/(2L^2)$, $[\ell^2]d_3 = (\Psi''(\theta) - L)/(8L^3)$, and $[\ell^{n-1}]d_n = (-1)^{n-1}\Psi^{(n-1)}(\theta)/(2^{n-1}L^n(n-1)!)$ for $n \geq 4$. For the logarithmic coefficients, $\deg_\ell h_1 = \deg_\ell h_2 = 1$, $\deg_\ell h_n \leq n - 1$ ($n \geq 3$), with $[\ell]h_1 = \frac{1}{2}$, $[\ell]h_2 = (\Psi'(\theta) - 1)/(2L)$, $[\ell^2]h_3 = (L - \Psi''(\theta))/(8L^2)$, and $[\ell^{n-1}]h_n = (-1)^n\Psi^{(n-1)}(\theta)/(2^{n-1}L^{n-1}(n-1)!)$ for $n \geq 4$. Finally $\deg_\ell g_n \leq n$ and the multiplicative top logarithm is universal:

$$\boxed{[\ell^n]g_n = \frac{1}{2^n n!}}, \quad g_n = \frac{\ell^n}{2^n n!} + \mathcal{O}_n((1 + |\ell|)^{n-1}) \text{ uniformly in } \theta. \quad (4.3)$$

Proof. In (2.1) a product Φ^k has ℓ -degree at most $k \leq n$, and for $k = n$ the coefficient requested is $[w^{n-1}]\Phi(0, w)^n/n$, whose nominal ℓ^n term carries no w and dies for $n \geq 2$; the $\ell^{n-1}w^{n-1}$ term takes $a\ell$ from $n - 1$ factors and ϕ from one, the multiplicity n cancelling $1/n$ — this is (4.2), and expanding ϕ gives the special cases. In $H = \log(1 + \beta z + zD) - LD$ the logarithm at order z^n involves only $d_{<n}$ (degree $\leq n - 2$ for $n \geq 3$), so the top logarithm of h_n is $-L[\ell^{n-1}]d_n$; orders 1, 2 are read off (3.2). For g_n , the only way to reach ℓ -degree n at z -weight n in $\exp(H)$ is $(h_1 z)^n/n!$ with the term $\ell/2$ chosen in every factor: $[\ell^n]g_n = (1/2)^n/n!$. (Equivalently, in the direct d -route: only $(-LD)^n/n!$ with $d_1 = -\ell/(2L) + \mathcal{O}(1)$ from every copy attains degree n , giving $(-L)^n(-1/2L)^n/n!$ — the two absorbed proofs of the same law.) $\square$

Corollary 4.2 (First subleading multiplicative logarithm). Write $h_1 = \frac{1}{2}\ell + q(\theta)$, $q = \kappa_0 - \Psi(\theta)$, and $\alpha_j(\theta) = [\ell^{j-1}]h_j$ ($j \geq 2$). Then for $n \geq 2$

$$[\ell^{n-1}]g_n = \frac{q(\theta)}{2^{n-1}(n-1)!} + \sum_{j=2}^n \frac{\alpha_j(\theta)}{2^{n-j}(n-j)!} : \quad (4.4)$$

the first nonuniversal logarithmic layer is closed-form in the top-log sequence of the h_j .

The universal law explains why elementary Lambert approximations look deceptively good — the largest slow logarithms exponentiate as if the periodic term were absent — while the phase data, entering one degree lower, are of the same size as everything else at that degree and can never be dropped.

4.2 Transfer of the newest periodic derivative

Theorem 4.3 (Inverse top-jet law). Viewing each coefficient as a differential polynomial in Ψ , for every $n \geq 1$ the monomial linear in the newest jet $\Psi^{(2n-2)}$ has

$$[\Psi^{(2n-2)}]d_n = \frac{(-1)^{n-1}}{2^{n-1}(n-1)!L^{2n-1}}, \quad [\Psi^{(2n-2)}]h_n = [\Psi^{(2n-2)}]g_n = \frac{(-1)^n}{2^{n-1}(n-1)!L^{2n-2}}. \quad (4.5)$$

Proof. For $n \geq 2$ the unique source of jet order $2n-2$ in (2.1) is the $k = 1$ term through $A_{n-1}(\theta)z^{n-1}/L$: a factor $A_j(\theta + w)z^j$ in a term of Lagrange index k has $j \leq n - k$ and gains at most $k - 1$ derivatives from the w -extraction, so its maximal jet order $2j + k - 1 \leq 2n - k - 1$ reaches $2n - 2$ only at $(k, j) = (1, n - 1)$; pure Ψ -factors rank lower. The forward top-jet law (1.4) then gives the d -statement. At order n the logarithm in H uses only $d_{<n}$ (jets $\leq 2n - 4$), so h_n inherits $-L$ times the d_n coefficient, and Bell products of lower h_j leave g_n with the same newest-jet coefficient. $\square$

At each order a genuinely new derivative $\Psi^{(2n-2)}$ enters linearly with a nonzero universal coefficient: the periodic hierarchy never closes at finite jet order, and an omission or sign error in A_{n-1} is immediately visible in h_n, g_n — a practical computer-algebra diagnostic.

5 Fourier-mode algebra and the analyticity strip

Proposition 5.1 (Finite Gamma–zeta convolutions). *For a finite jet monomial $\mathcal{M} = \prod_r (\Psi^{(r)})^{m_r}$,*

$$\widehat{\mathcal{M}}(k) = \sum_{\substack{k_{r,s} \in \mathbb{Z} \setminus \{0\} \\ \sum k_{r,s} = k}} \prod_r \prod_{s=1}^{m_r} (2\pi i k_{r,s})^r \widehat{\Psi}(k_{r,s}), \quad (5.1)$$

and every Fourier coefficient of every d_n, h_n, g_n is a finite linear combination of such convolutions with coefficients polynomial in ℓ and rational in L, β .

Proposition 5.2 (Common strip). *For every $\sigma < \pi/(2L)$, all periodic coefficient functions of all d_n, h_n, g_n extend holomorphically and boundedly to $|\Im \theta| \leq \sigma$, with normally convergent Fourier series: Stirling on vertical lines gives $|\widehat{\Psi}(k)| \leq C_\varepsilon \exp[-(\pi^2/L - \varepsilon)|k|]$, and differentiation, multiplication, and composition preserve the exponential type. Consequently, truncating all Fourier sums at $|k| \leq K$ costs at most $C(1 + \ell)^N K^{\mathcal{O}(1)} e^{-(\pi^2/L - 2\pi\sigma)K}$ at fixed inverse order N — a few modes suffice for high-precision real-phase evaluation, since already $|\widehat{\Psi}(1)| = 1.06271 \dots \times 10^{-6}$.*

6 Phase locking and coefficient tomography

The fast phase can be frozen exactly while the endpoint is approached, turning the formal coefficient functions into observable limits.

Theorem 6.1 (Phase-locked endpoint sequences). *Fix $u \in [0, 1)$ and set $\rho_{m,u} = m + u - \beta$, $y_{m,u} = \exp(-\frac{L}{2}\rho_{m,u}^2)$ for large integers m . Then $\theta \equiv u \pmod{1}$ along the sequence, and (1.11) holds with $(\ell, \theta) = (\log \rho_{m,u}, u)$, uniformly in u . In particular, the limit set of the first normalized correction $\rho(G/G_0 - 1) - \frac{1}{2}\ell$ is exactly the interval $[\kappa_0 - \max \Psi, \kappa_0 - \min \Psi]$, every point being realized by a phase-locked sequence.*

Writing $g_n(\ell, u) = \sum_{r \leq n} g_{n,r}(u) \ell^r$, the renormalized statistic

$$\mathcal{T}_{N,m}(u) = \rho_{m,u}^N \left\{ \frac{G(y_{m,u})}{G_0(y_{m,u})} - 1 - \sum_{n < N} \frac{g_n(\log \rho_{m,u}, u)}{\rho_{m,u}^n} \right\} = g_N(\log \rho_{m,u}, u) + \mathcal{O}_N \left(\frac{(1 + \log \rho_{m,u})^{N+1}}{\rho_{m,u}} \right) \quad (6.1)$$

reconstructs, phase by phase, every periodic layer $g_{N,r}(u)$ from a polynomial fit in $\log \rho$; the universal $g_{N,N} = 1/(2^N N!)$ supplies an exact normalization. The endpoint has no single asymptotic constant beyond the elementary factor: it has a hierarchy of periodic coefficient *profiles*, whose Fourier spectra (5.1) are predicted explicitly.

7 Elasticity and every inverse derivative

Since $T = L\rho^2/2$ and $y = e^{-T}$, the endpoint Euler operator acts through the two-scale derivation

$$y \frac{d}{dy} = -\frac{1}{L\rho} \mathfrak{d}, \quad \mathfrak{d} = \partial_\theta + z \partial_\ell - z^2 \partial_z, \quad \mathfrak{d}(f z^n) = (\partial_\theta f) z^n + (\partial_\ell f - n f) z^{n+1}. \quad (7.1)$$

Theorem 7.1 (All-orders elasticity). *The elasticity $\mathcal{E}(y) = yG'(y)/G(y)$ has $\mathcal{E} = \sum_{m \geq 1} e_m z^m$ with $e_1 = 1$ and, for $m \geq 2$ (with $h_0 := 0$),*

$$e_m = -\frac{\delta_{m,2}}{L} - \frac{1}{L} \left(\partial_\theta h_{m-1} + \partial_\ell h_{m-2} - (m-2)h_{m-2} \right); \quad \mathcal{E} = \frac{1}{\rho} + \frac{\Psi'(\theta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{1+\ell}{\rho^3}\right). \quad (7.2)$$

Proof. Apply $-(L\rho)^{-1}\mathfrak{d}$ to $\log G = \ell - L\rho - L\beta + \sum h_n \rho^{-n}$: the elementary part contributes $z - z^2/L$, and (7.1) termwise gives the recurrence; $\partial_\theta h_1 = -\Psi'$ gives the two-term form, recovering the corpus's theorem. $\square$

Two complementary routes organize *all* higher derivatives – one recursive in the derivative order, one closed via Stirling–Bell conversion.

Theorem 7.2 (Derivative hierarchy and the universal two-term law). *The ratios $Q_m = y^m G^{(m)}/G$ obey the exact recurrence $Q_{m+1} = y \frac{d}{dy} Q_m + (\mathcal{E} - m)Q_m$ (linear at fixed $\mathcal{E}$), whose coefficient form via (7.1) computes every fixed $G^{(m)}$ to arbitrary order. Its two-term consequence is universal: for each fixed $m \geq 1$,*

$$G^{(m)}(y) = (-1)^{m-1} (m-1)! \frac{G(y)}{y^m \rho} \left[1 + \frac{(\Psi'(\theta) - 1)/L - H_{m-1}}{\rho} + \mathcal{O}_m\left(\frac{1+\ell}{\rho^2}\right) \right], \quad (7.3)$$

so the quantile is eventually increasing, concave, with alternating higher derivatives near the endpoint (each fixed m on its own interval $0 < y < y_m$; no uniformity in m is asserted). Alternatively, with $E = y \, d/dy$, $f = \log G$, and signed Stirling numbers $s(k, j)$,

$$H_k := y^k f^{(k)}(y) = \sum_{j=1}^k s(k, j) E^j f, \quad \frac{y^r G^{(r)}(y)}{G(y)} = Y_r(H_1, H_2, \dots, H_r), \quad (7.4)$$

a closed finite algorithm assembling every ordinary derivative from repeated applications of (7.1) to $\log G$.

Proof. The recurrence is the product rule on $G^{(m+1)} = (G^{(m)})'$. Writing $Q_m = a_m z(1 + b_m z + \dots)$, the operator $y \, d/dy$ raises the displayed order, so $a_{m+1} = -ma_m$ and $b_{m+1} = b_m - 1/m$; from $Q_1 = \mathcal{E}$, $a_m = (-1)^{m-1} (m-1)!$ and $b_m = e_2 - H_{m-1}$ with $e_2 = (\Psi'(\theta) - 1)/L$. The Stirling–Bell route is the classical Euler-to-ordinary conversion followed by the complete-Bell exponential formula for $G = e^f$. $\square$

8 The forward saddle engine: constrained partition sums

The inverse formulas consume the periodic forward coefficients A_j . This chapter closes their construction with a finite partition formula for the universal Gaussian layer of the saddle analysis of the Bromwich integral.

Let $s > 0$ solve the saddle equation $x = -K'(s)$ and normalize the saddle jets and their ratios

$$\varkappa_r = s^r K^{(r)}(s), \quad \varrho_r = \frac{\varkappa_r}{\varkappa_2} \quad (r \geq 3). \quad (8.1)$$

Scaling the contour by $t = s(1 + iZ/\sqrt{\varkappa_2})$, $Z \sim N(0, 1)$, produces the formal correction factor

$$\mathcal{S} = \mathbb{E} \left[\left(1 + \frac{iZ}{\sqrt{\varkappa_2}} \right)^{-1} \exp \left(\sum_{r \geq 3} \frac{\varrho_r}{r!} \frac{(iZ)^r}{\varkappa_2^{(r-2)/2}} \right) \right] \sim 1 + \sum_{n \geq 1} \frac{C_n}{\varkappa_2^n}, \quad (8.2)$$

the reciprocal factor being the CDF's Bromwich $1/t$ – deleting it would produce the *density* coefficients instead.

Theorem 8.1 (Constrained-partition formula). *For $n \geq 1$,*

$$C_n = \sum_{\substack{m_0, m_3, m_4, \dots \geq 0 \\ m_0 + \sum_{r \geq 3} (r-2)m_r = 2n}} (-1)^{m_0 + N/2} (N-1)!! \prod_{r \geq 3} \frac{\varrho_r^{m_r}}{m_r! (r!)^{m_r}}, \quad N := m_0 + \sum_{r \geq 3} r m_r, \quad (8.3)$$

a finite sum (the constraint forces N even and $r \leq 2n+2$). When all ϱ_r vanish it reduces to the Gaussian Mills-ratio coefficients $C_n = (-1)^n (2n-1)!!$ – a sensitive sign and normalization check, verified symbolically $(-1, 3, -15)$. The logarithmic coefficients $\log \mathcal{S} \sim \sum \Lambda_n \varkappa_2^{-n}$ follow nonrecursively from

$$\Lambda_n = \sum_{k=1}^n \frac{(-1)^{k-1}}{k} \sum_{n_1 + \dots + n_k = n, n_j \geq 1} C_{n_1} \cdots C_{n_k}, \quad \Lambda_1 = C_1, \quad \Lambda_2 = C_2 - \frac{1}{2} C_1^2. \quad (8.4)$$

Proof. Expand the reciprocal geometrically (m_0 factors of $-iZ/\sqrt{\varkappa_2}$) and the exponential by multiplicities m_r . The total inverse half-power of $\varkappa_2$ is $\frac{1}{2}(m_0 + \sum (r-2)m_r)$, giving the constraint; the Gaussian degree is N with $N-2n = 2 \sum m_r$ even, so $\mathbb{E}Z^N = (N-1)!!$; the powers of i and the reciprocal's sign give $(-1)^{m_0 + N/2}$; and the exponential series supplies the multiplicity factorials. The Λ formula is $\log(1 + \cdot)$ composed with the C -series. $\square$

The first coefficients are $C_1 = -1 - \frac{1}{2}\varrho_3 + \frac{1}{8}\varrho_4 - \frac{5}{24}\varrho_3^2$ and

$$C_2 = 3 + \frac{5}{2}\varrho_3 - \frac{5}{8}\varrho_4 + \frac{1}{8}\varrho_5 - \frac{1}{48}\varrho_6 + \frac{35}{24}\varrho_3^2 - \frac{35}{48}\varrho_3\varrho_4 + \frac{7}{48}\varrho_3\varrho_5 + \frac{35}{384}\varrho_4^2 + \frac{35}{48}\varrho_3^3 - \frac{35}{64}\varrho_3^2\varrho_4 + \frac{385}{1152}\varrho_3^4. \quad (8.5)$$

With $q = \log_2 s$ and $J(q) = K(2^q)$, the normalized jets are generated exactly by the operator identity $\varkappa_r(q) = L^{-r} \prod_{j=0}^{r-1} (D_q - jL) J(q)$; inserting the exact decomposition of K from [section 9](#) and converting from the saddle coordinate q to the exact Lambert phase completes the pipeline

$$\Psi \longrightarrow \{\varkappa_r\} \xrightarrow{(8.3)} \{C_n\} \xrightarrow{(8.4)} \{\Lambda_n\} \longrightarrow \{A_j\} \xrightarrow{(2.1)} \{d_n\} \xrightarrow{(2.3)} \{g_n\}, \quad (8.6)$$

every arrow a finite differential-polynomial or coefficient-extraction operation at fixed order. The raw term count of (8.3) grows like an integer-partition function – subexponentially – rather than like all set partitions, which removes the practical bottleneck to generating $A_3, A_4, \dots$ in compact Fourier-differential form.

9 The exact 2-adic completion of the Laplace exponent

Everything so far is asymptotic in ρ . This chapter is *exact*: it completes the quadratic-periodic normal form of K by an explicitly 2-adic, everywhere-convergent tail, and shows that the periodic phase and the beyond-all-orders sectors are two faces of one Mellin kernel.

9.1 The decomposition and the dyadic Bose tail

The product (1.1) gives the dyadic difference equation $K(2t) - K(t) = \log(1 - e^{-t}) - \log t$. Define the *dyadic Bose tail*

$$B(t) = - \sum_{k \geq 0} \log(1 - e^{-2^k t}), \quad B(t) = - \log(1 - e^{-t}) + B(2t). \quad (9.1)$$

Theorem 9.1 (Exact quadratic-periodic-dyadic decomposition). *For every $t > 0$,*

$$K(t) = - \frac{\log^2 t}{2L} + \frac{1}{2} \log t - C_B + \Psi(\log_2 t) + B(t). \quad (9.2)$$

Proof. With $Q(q) = -\frac{L}{2}q^2 + \frac{L}{2}q$ one checks $Q(q+1) - Q(q) = -Lq$, so $\Omega(q) := K(2^q) - Q(q) - B(2^q)$ is 1-periodic by the two difference equations. The small- t Mellin expansion of B (theorem 9.3 below) is $B(t) = \log^2(1/t)/(2L) + \frac{1}{2}\log(1/t) + C_B - \Psi(\log_2 t) + o(1)$, and $K(t) \rightarrow 0$ as $t \downarrow 0$; letting $q = m + u \rightarrow -\infty$ at fixed u identifies $\Omega(u) = \Psi(u) - C_B$, and periodicity upgrades the identity to all q . $\square$

Theorem 9.2 (Exact 2-adic coefficients). *For every $t > 0$,*

$$B(t) = \sum_{n \geq 1} b_n e^{-nt}, \quad b_n = \frac{2^{\nu_2(n)+1} - 1}{n}, \quad 0 < b_n < 2, \quad (9.3)$$

with the elementary tail enclosure $0 < B(t) - \sum_{n \leq N} b_n e^{-nt} \leq 2e^{-(N+1)t}/(1 - e^{-t})$.

Proof. Expand each Bose logarithm and regroup $B(t) = \sum_{k \geq 0} \sum_{m \geq 1} m^{-1} e^{-m2^k t}$: the representations $n = m2^k$ are indexed by $0 \leq k \leq \nu_2(n)$, with total weight $\frac{1}{n} \sum_{k \leq \nu_2(n)} 2^k = (2^{\nu_2(n)+1} - 1)/n$. $\square$

The first terms, $B(t) = e^{-t} + \frac{3}{2}e^{-2t} + \frac{1}{3}e^{-3t} + \frac{7}{4}e^{-4t} + \frac{1}{5}e^{-5t} + \frac{1}{2}e^{-6t} + \dots$, spike at powers of two: the exact arithmetic trace of the dyadic scale hierarchy. The integer weight $f(n) = nb_n = 2^{\nu_2(n)+1} - 1$ satisfies $f(2n) = 2f(n) + 1$, $f(2n+1) = 1$, and its generating function obeys the Mahler equation $C(z) = 2C(z^2) + z/(1-z)$, placing the sectors in the same digital universe as the Thue–Morse and ruler sequences – but with geometric accumulation over the full 2-adic valuation in place of digit-sum parity. Note the crosswalk: $\nu_2(n) + 1$ is precisely the integer zero order of $\widehat{\text{up}}$ that drives the exactness theorems of the companion volume *Dyadic-Comb Frontiers*; here the same statistic returns as the arithmetic seed of the beyond-all-orders sectors.

9.2 One Mellin kernel, two asymptotic ends

Theorem 9.3 (Mellin bridge). *For $\Re s > 0$,*

$$\sum_{n \geq 1} \frac{b_n}{n^s} = \frac{\zeta(s+1)}{1 - 2^{-s}}, \quad \int_0^\infty B(t) t^{s-1} dt = \Gamma(s) \frac{\zeta(s+1)}{1 - 2^{-s}}, \quad (9.4)$$

and contour shifting in the Mellin inversion gives, for every fixed M , with $R = \log(1/t)$,

$$B(t) = \frac{R^2}{2L} + \frac{R}{2} + C_B - \Psi(\log_2 t) + \sum_{m=1}^{M-1} \frac{(-1)^m \zeta(1-m)}{m! (1 - 2^{-m})} t^m + \mathcal{O}_M(t^M) \quad (t \downarrow 0). \quad (9.5)$$

Proof. The Dirichlet series is the double sum $\sum_k \sum_m m^{-1} (m2^k)^{-s} = \zeta(s+1) \sum_k 2^{-ks}$; termwise Mellin transformation of (9.3) gives the transform. Shifting the inversion contour left: the order-three pole at $s = 0$ of $\Gamma(s)\zeta(1+s)(1 - 2^{-s})^{-1}$ yields the quadratic polynomial and C_B ; the simple poles at $s = \chi_k$ yield exactly $-\Psi(\log_2 t)$ by (1.3); and the poles of Γ at $-m$ yield the Bernoulli-type powers. $\square$

Two faces of one kernel

The Gamma–zeta phase Ψ (small- t residues at the complex dimensions $2\pi i k/L$) and the exact 2-adic exponential sectors (large- t Dirichlet regrouping) are generated by the *same* kernel $\Gamma(s)\zeta(s+1)/(1 - 2^{-s})$. No new frequencies ever appear downstream: the entire hierarchy of inverse oscillations consists of integer convolutions of the modes χ_k (theorem 5.1), and the beyond-all-orders remainder is not an opaque error but the explicit arithmetic object (9.3). As a byproduct, comparing the two derivations of the constant term proves $C_\# + C_B = -\frac{1}{2} \log(2\pi)$.

9.3 Exact saddle map and ultraflat jets

Corollary 9.4 (Exact 2-adic saddle map). *With $t = 2^q$, the saddle abscissa $x = -K'(t)$ is exactly*

$$x(q) = 2^{-q} \left(q - \frac{1}{2} - \frac{\Psi'(q)}{L} \right) + \sum_{n \geq 1} (2^{\nu_2(n)+1} - 1) e^{-n2^q}, \quad (9.6)$$

the first term the complete algebraic–periodic saddle map, the second an exact convergent ultraflat correction. Derivative bookkeeping is closed: with Stirling numbers of the second kind,

$$\frac{d^r}{dq^r} B(2^q) = L^r \sum_{n \geq 1} b_n \sum_{j=0}^r \left\{ \begin{matrix} r \\ j \end{matrix} \right\} (-n2^q)^j e^{-n2^q}, \quad t^r B^{(r)}(t) = (-1)^r \sum_{n \geq 1} b_n (nt)^r e^{-nt}. \quad (9.7)$$

The jet formulas expose the technical obstacle to naive hyperasymptotics: each q -derivative of an e^{-n2^q} sector produces powers of 2^q , so the ordinary inverse-power saddle hierarchy is *not uniform* inside a fixed dyadic sector; and although the algebraic inversion gives the saddle coordinate $q_*(y) = \theta + \mathcal{O}(\ell/\rho)$, the induced change in the exponent $n2^{q_*}$ is of order $2^\theta \ell/\rho \rightarrow \infty$ – the exact saddle coordinate is part of the beyond-all-orders datum ([theorem 12.5](#)).

10 Verification

The two absorbed packages verify overlapping claims by independent implementations; both suites live under `assets/`.

Symbolic. Over algebraically independent symbols $L, \beta, C_\sharp, \ell, P_0, \dots, P_4, A_1, A'_1, A_2$, both codes confirm $d_j^{\text{diagonal}} - d_j^{\text{triangular}} \equiv 0$ ($j \leq 3$), the third correction (3.1), the universal top-log (4.3), and the top-jet transfer (4.5); the saddle generator reproduces $C_n = (-1)^n (2n-1)!!$ at vanishing ϱ and archives C_1, C_2, C_3, Λ_2 (`generated_coefficients.tex`).

Exact identities. At 70–100 digits, the decomposition (9.2) agrees with the defining product for K to $\sim 10^{-16}$ at $t \in \{0.1, 1, 3, 10\}$ (twelve Fourier modes); the 2-adic series (9.3) agrees with the product to 7.5×10^{-39} at $t = 0.7$ and 3.8×10^{-55} at $t = 1.5$; the small- t expansion (9.5) to $\sim 4 \times 10^{-16}$ on $0.02 \leq t \leq 0.2$; and the Dirichlet identity (9.4) to the expected algebraic truncation tail with 5×10^4 unaccelerated terms.

Exact dyadic quantiles. With $a_0 = 1$ and $a_n = (2^n - 1)^{-1} \sum_{k < n} a_k / (n - k + 1)!$, the repository’s exact values give $y_n = F(2^{-n}) = 2^{-n(n-1)/2} a_n$ with $G(y_n) = 2^{-n}$ *exactly* – no inversion error, only the asymptotic formula is evaluated (at 100 digits, twelve modes). Signed relative errors of $G^{[r]} = G_0 \exp(\sum_{j \leq r} h_j \rho^{-j})$:

n	ρ	G_0 only	through h_1	through h_2	through h_3
5	6.44	-3.83×10^{-1}	7.89×10^{-2}	-2.17×10^{-2}	6.41×10^{-3}
10	12.08	-2.57×10^{-1}	2.69×10^{-2}	-4.40×10^{-3}	7.19×10^{-4}
20	22.82	-1.62×10^{-1}	8.76×10^{-3}	-8.24×10^{-4}	7.40×10^{-5}
40	43.65	-9.66×10^{-2}	2.72×10^{-3}	-1.43×10^{-4}	6.96×10^{-6}
80	84.54	-5.53×10^{-2}	8.05×10^{-4}	-2.34×10^{-5}	6.05×10^{-7}
120	125.08	-3.94×10^{-2}	3.88×10^{-4}	-7.92×10^{-6}	1.40×10^{-7}
160	165.47	-3.08×10^{-2}	2.30×10^{-4}	-3.64×10^{-6}	4.92×10^{-8}

Table 1: Exact-quantile tests: the new third correction improves the two-correction error by factors 11–74 across the tested range.

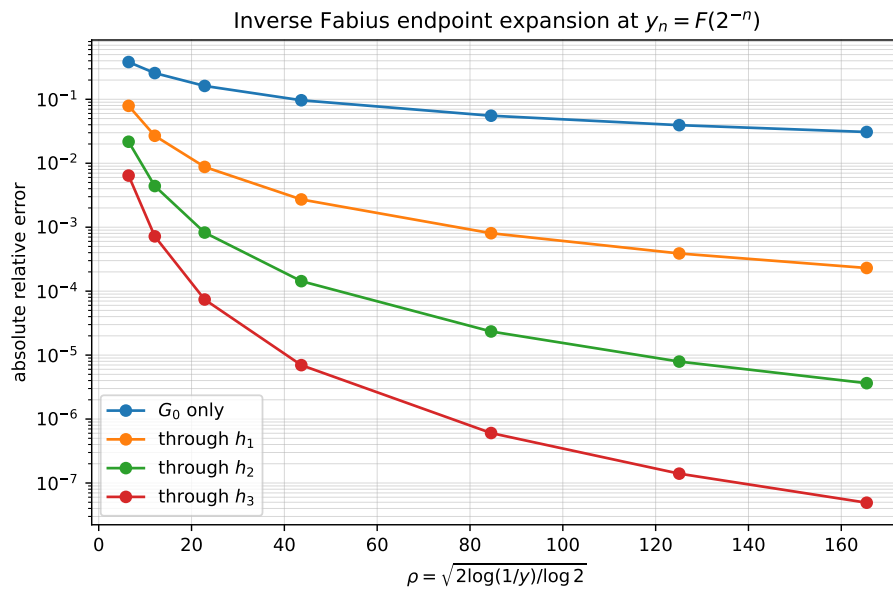


Figure 1: Absolute relative errors on the exact dyadic quantiles; each added logarithmic correction produces the expected strong reduction once ρ is moderately large.

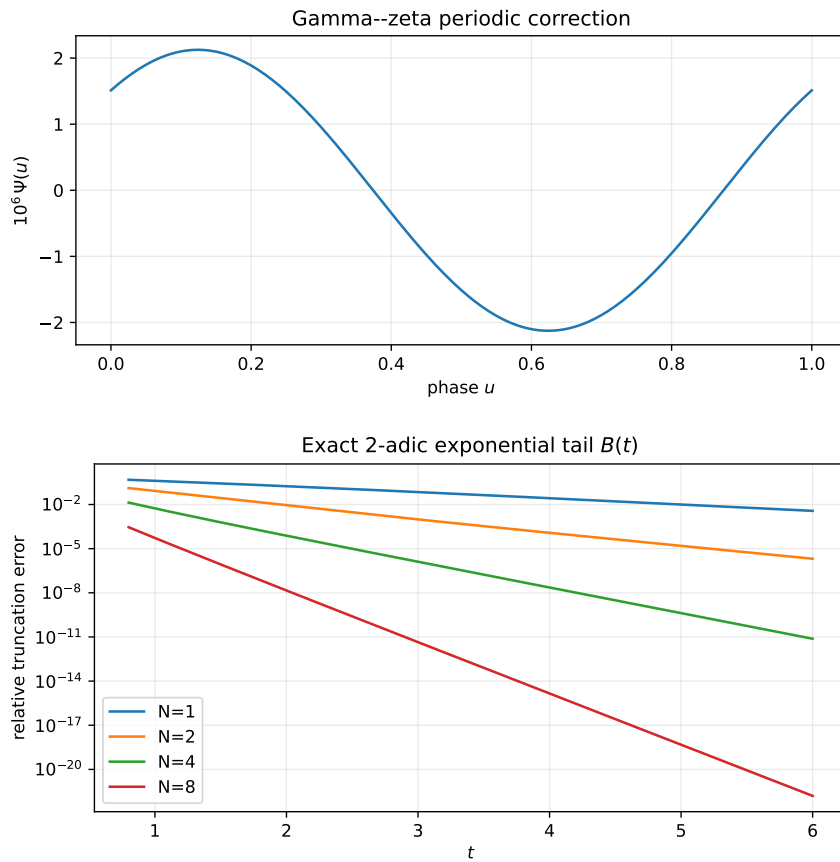


Figure 2: Top: the centered Gamma-zeta fluctuation Ψ – tiny amplitude, same asymptotic order as the constants. Bottom: relative error of $N = 1, 2, 4, 8$ terms of the exact 2-adic series for $B(t)$, with the elementary enclosure of [theorem 9.2](#).

11 Connections across the corpus

Literal small- y form. Substituting $\rho = \sqrt{2T/L}$ back, the complete expansion reads

$$G(y) = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}} \exp\left\{\sum_{n=1}^N \left(\frac{L}{2T}\right)^{n/2} h_n\left(\frac{1}{2} \log \frac{2T}{L}, \sqrt{2T/L} + \beta\right)\right\} \left[1 + \mathcal{O}_N\left(\frac{(1 + \log T)^N}{T^{(N+1)/2}}\right)\right]. \quad (11.1)$$

Lambert geometry. The method performs the saddle inversion in λ and invokes the exact Lambert identity only once, through (1.12): that is why the coefficients are polynomial in the single logarithm ℓ rather than an uncontrolled nest, and why β (which completes the square in the dominant exponent) is the only constant shift. *Bell structures.* Two Bell mechanisms compose without flattening: the forward A_j are Bell assemblies of saddle jets, the inverse g_n are Bell assemblies of the h_j — the diagonal formula is the “Bell-of-Bell” transform with a diagonal grading. *Thue–Morse and q -data.* The exact dyadic quantiles that verify the expansion are the same q -binomial arithmetic that runs the companion volumes; the 2-adic seed b_n shares its statistic $\nu_2(n) + 1$ with the sinc zero orders; and the up-function reading of the expansion is geometric: it is the all-orders law of the inward displacement from the support boundary of the Rvachev bell at prescribed tiny height, the Gamma–zeta oscillation being the Mellin shadow of the same dyadic lattice that generates the sinc product.

12 Conjectures and frontier directions

Conjecture 12.1 (Gevrey-one growth). *On compact real phase sets (and in weighted analytic norms on a strip, with weight $1 + |\ell|$ for the polynomial part), $|h_n| + |g_n| \leq C A^n n!$. The top-jet transfer (4.5) combined with a Cauchy estimate at the strip boundary $\pi/(2L)$ gives the heuristic exponential constant $8/\pi^2$ — a Gevrey-one, not worse, divergence — but other saddle diagrams may reinforce or partially cancel it.*

Conjecture 12.2 (Optimal truncation). *At fixed real phase, truncation near $N_{\text{opt}} \sim (\pi^2/8)\rho$ leaves an exponentially small remainder whose leading exponential and Stokes data are governed by the nearest complex singularities of Ψ and the lower-Lambert branch geometry.*

Conjecture 12.3 (Sharp strip inheritance). *Each periodic coefficient of d_n, h_n, g_n extends to the same maximal strip as Ψ and no generic cancellation enlarges it: the Fourier decay rate is stable under all inverse operations.*

Conjecture 12.4 (Nonconstant phase at every order). *Every d_n, h_n, g_n has a nonconstant periodic component for generic ℓ : no finite truncation beyond the elementary factor is phase-blind. (The top-jet law makes this overwhelmingly plausible; a certified proof could isolate the $|k| = 1$ modes against the enormous decay gap to $|k| \geq 2$.) At higher orders the renormalized phase profiles trace closed analytic curves; for generic jet selections these should be immersed, and for rich selections embedded, with embedding dimension controlled by the first Gamma–zeta modes — a “quantile cluster geometry.”*

Conjecture 12.5 (Two-level exponentially improved transseries). *After optimal truncation, $G(y)$ admits an exponentially improved description with at least two distinct levels: saddle-resurgent sectors controlled by the large-order growth of the A_j , and ultraflat dyadic sectors $\exp(-n2^{q_*(y)})$ seeded by $b_n = (2^{\nu_2(n)+1} - 1)/n$, where q_* is the exact saddle coordinate of (9.6) (not replaceable by θ ; see section 9). Within the ultraflat level, the Stokes/connection data should be Dirichlet convolutions of b_n , i.e. finite products of $\zeta(s+1)/(1-2^{-s})$ and shifts — a precise, computationally testable resurgent-arithmetic prediction.*

Conjecture 12.6 (Universal diagonal inversion class). *Whenever a distribution has a forward exact-phase expansion $-a\lambda^2 + b\lambda - c \log \lambda + C + P(\lambda) + \sum A_j(\lambda)\lambda^{-j}$ with periodic analytic P, A_j , its*

inverse admits the entire package proved here — diagonal expansion with polynomial logarithms, phase locking, newest-jet transfer, Bell conversion — with universal highest logarithms depending only on a and c . For base $b > 1$ smoothing laws the 2-adic tail generalizes with coefficients $(b^{\nu_b(n)+1} - 1)/((b-1)n)$ and kernel $\zeta(s+1)/(1-b^{-s})$, the phase acquiring period one in $\log_b$.

Concrete projects: generate A_3, A_4 compactly through the pipeline (8.6) and make g_4, g_5 explicit in Ψ alone; prove weighted Wiener-algebra bounds uniform in phase; run optimal truncation experiments to see whether the first visible beyond-all-orders scale is saddle-resurgent or the far smaller dyadic one; quantify the uniform-in- m breakdown of the derivative law (7.3); locate the nearest complex critical values of F (the analytic continuation radius of G and the Borel singularities); and build certified few-mode Fourier-polynomial coefficient tables $g_n \approx \sum_{r \leq n} \sum_{|k| \leq K} c_{n,r,k} \ell^r e^{2\pi i k \theta}$ that, with one Newton step on the exact functional equation, give a fast globally certified quantile algorithm.

Lean formalization roadmap

Four layers, the first three purely formal: (1) a two-scale coefficient ring (truncated series over $\mathbb{R}[\ell] \otimes C^\infty(\mathbb{R}/\mathbb{Z})$, closure under composition, log, exp); (2) parameterized Lagrange inversion $[t^k]W = \frac{1}{k}[w^{k-1}]\Phi^k$ and the diagonal specialization (2.1); (3) the Gaussian contraction (8.3) over an algebraic moment functional, and the 2-adic regrouping and Dirichlet identities of section 9 (a bijection and two Tonelli arguments); (4) the analytic transfer (section A) from formal cancellation to the fixed-order remainder. The degree/top-jet filtrations of section 4 are finite combinatorics on top of (2).

A Proof details

Finiteness bookkeeping. In (2.1) the index- k summand requests $z^{n-k}w^{k-1}$: no Φ -power beyond z^{n-k} , no Taylor jet beyond order $k-1$, and A_j only for $j \leq n-k$; hence A_{n-1} occurs only at $k=1$ — the key to the top-jet proof.

Formal-to-asymptotic transfer. With $\mathcal{F}(w) = \frac{L}{z}w + \mathcal{R}_N(w) + E_N(w)$, $E_N = \mathcal{O}_N(z^N)$, formal cancellation gives $\mathcal{F}(D^{[N]}) = \mathcal{O}_N((1+\ell)^N z^N)$, while $\inf |\partial_w \mathcal{F}| \geq L/(2z)$ throughout the relevant $\mathcal{O}(\ell z)$ neighborhood for small z ; hence $|D - D^{[N]}| \leq (2z/L) |\mathcal{F}(D^{[N]})| = \mathcal{O}_N((1+\ell)^N z^{N+1})$ — the gain-of-one-power argument that lets a forward expansion through A_{N-1} certify an inverse expansion through d_N . The map $\lambda \mapsto \lambda e^{-L\lambda}$ then transfers phase error to relative inverse error of the same order.

B Coefficient dictionary

$$\begin{aligned} L &= \log 2, & \beta &= \frac{1}{L} + \frac{1}{2}, & T &= \log(1/y), & \rho &= \sqrt{2T/L}, \\ z &= \rho^{-1}, & \ell &= \log \rho, & \theta &= \rho + \beta, & G_0 &= \rho e^{-L\rho - L\beta}, \\ \lambda &= \rho + \beta + D, & D &= \sum d_n z^n, & d_n &= \sum_{k=1}^n \frac{(-1)^k}{kL^k} [z^{n-k}w^{k-1}]\mathcal{R}^k, & \frac{G}{G_0} &= (1 + \beta z + zD)e^{-LD}, \end{aligned}$$

with $\mathcal{R}$ from (1.9); these lines suffice to implement the complete inverse expansion once the forward A_j are available (section 8). In evaluation: preserve the exact phase $\theta = \rho + \beta$ inside every periodic coefficient (replacing it by ρ shifts the first oscillatory correction); work through $\log G$ at extreme y ; and use the Fourier form of Ψ , whose differentiated modes keep exponential decay. For reference,

$$\Lambda_2 = \frac{5}{16}\varrho_3^4 + \frac{5}{8}\varrho_3^3 - \frac{25}{48}\varrho_3^2\varrho_4 + \frac{9}{8}\varrho_3^2 - \frac{2}{3}\varrho_3\varrho_4 + \frac{7}{48}\varrho_3\varrho_5 + 2\varrho_3 + \frac{1}{12}\varrho_4^2 - \frac{1}{2}\varrho_4 + \frac{1}{8}\varrho_5 - \frac{1}{48}\varrho_6 + \frac{5}{2}. \quad (\text{B.1})$$

C Provenance of the absorbed sources

Source (directory; SHA-256 of main .tex)	Contribution
<i>Closed All-Orders Endpoint Asymptotics for the Inverse Fabius Function</i> (inverse_fabius_all_orders_package; SHA in assets/SHA256SUMS-absorbed.txt)	Diagonal formulas including the general composition and the h_n/g_n diagonal forms; fixed-order theorem with logarithmic remainder grading; degree and top-log laws; inverse top-jet law; A_2, A'_1 ; Fourier convolution algebra, analyticity strip, certified truncation; phase locking, cluster set, tomography; elasticity and the Q_m hierarchy with the harmonic-number law; exact-dyadic-quantile verification (table 1, figure 1); Gevrey/optimal-truncation/phase-geometry conjectures; proof-detail appendices.
<i>Complete Small-Argument Asymptotics of the Inverse Fabius Function</i> (inverse_fabius_asymptotics_report; SHA in assets/SHA256SUMS-absorbed.txt)	The closed d_n extractor (independent derivation) and the direct partial-Bell g_n formula; universal top-log (independent proof); third correction (independent, verified identical); the constrained-partition saddle engine, Mills specialization, Λ_n composition, jet operator, pipeline; the exact 2-adic completion: K -decomposition, dyadic Bose tail, Mahler and Dirichlet forms, Mellin bridge, $C_{\sharp} + C_B$ relation, exact saddle map, ultraflat jets; Stirling–Bell ordinary-derivative route; strip-inheritance/transseries/resurgent-arithmetic conjectures; formalization obligations; K/B /Dirichlet verification and the Ψ and tail-convergence figures.

What was deduplicated: the imported forward/inverse corpus inputs; the two-scale setup; the fixed-point reduction; the closed d_n formula (proved in both, one statement here); the universal top-log (two proofs, both retained in theorem 4.1); the third correction (independently derived, verified algebraically identical — the merge’s cross-validation); the Bell conversion; the elasticity recurrence (identical in both); and the audit/status scaffolding. Nothing mathematical was discarded; where the sources’ groupings differ, the compacter form is displayed and the other recorded in the adjacent remark. Exact SHA-256 hashes are in assets/SHA256SUMS-absorbed.txt.

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